

Eric Margay

Hardware & Software Engineer

Engineer with 7+ years building systems where code meets physical devices — from Python automation and Django services to firmware on ESP32, Raspberry Pi and Ultra-Wideband radio. Currently automating CAE PowerTrain simulation reporting at Ford's Global Technology & Business Center, and a long-standing workshop instructor in Python, IoT and web development.

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Professional Experience

Ford Motor Company · Global Technology & Business Center

FEB 2023 — PRESENT

Software Engineer, IPTSE CAE · Mexico City

- Build and maintain the Python automation that generates CAE PowerTrain virtual testing reports, replacing manual report assembly for the simulation teams.
- Develop Microsoft Power Apps workflows wired to Planner and SharePoint connectors to automate internal engineering processes.
- Design integrations between hardware product systems and internal engineering applications, unifying data across previously disconnected teams.
- Deliver custom Python and JavaScript tooling scoped directly with internal customers, from requirements through production release.
- Enforce PEP-8, clean-code and SOLID standards across the codebase through peer reviews and refactoring proposals.
- Document architecture and results in Jupyter Notebooks and Figma prototypes, and report weekly progress, blockers and findings to the team lead.

Python · JavaScript · Power Apps · SharePoint · Jupyter · Git · Figma · Agile

EUM Tech · Parking Tip

DEC 2021 — FEB 2023

Developer Engineer, Engineering Department · Mexico City

- Extended and integrated the company API across the full product suite, unifying hardware devices with the web applications that controlled them.
- Built a Python object-oriented sub-module driving smart signaling lighting over serial-port communication with the hardware firmware.
- Implemented real-time WebSocket channels between JavaScript clients and Python services to stream live device state.
- Shipped new MVC capabilities in Django spanning both back-end and front-end functionality.
- Administered remote servers over SSH, tuning Apache behaviour to validate data correlation for the Walmart account.
- Trained new hires on the company development process and provided direct technical support to customers on delivered work.

Python · Django · JavaScript · WebSockets · REST API · Apache · SSH · Figma

Lambda Mobility

MAY 2022 — 2024

Developer Engineer & Co-Founder · Mexico City

- Co-founded a real-time tracking venture and led both hardware and software development end to end.
- Engineered indoor positioning over Ultra-Wideband radio, implementing trilateration firmware in MicroPython and C++.
- Designed the electronic systems behind the tracking devices.
- Built the public-transport model application on Django MVC with WebSocket communication for live vehicle position.

MicroPython · C++ · UWB / RF · Django · WebSockets · PCB Design

Waltem Health

SEP 2019 — FEB 2020

Developer Engineer, Engineering Department · Mexico City

- Developed Python algorithms to process ECG sensor data for medical hardware.
- Built Raspberry Pi-to-cloud pipelines that pushed measurement payloads over HTTP requests.
- Reverse-engineered existing medical devices to enable an in-house hardware implementation.
- Consulted across departments to bridge software and hardware into a single solution for healthcare customers, documenting recurring issues and their fixes.

Python · Raspberry Pi · HTTP / REST · ECG / Sensors

Streamr Inc.

MAR 2019 — 2024

Freelance Engineer & Technology Evangelist · Switzerland (remote)

- Resolved integration issues between the Streamr API and IoT devices publishing to the blockchain data marketplace.
- Authored and delivered a Python crash course on Raspberry Pi and ESP32 boards, taking developers from raw sensor to live cloud stream.
- Structured and streamed sensor data to the Streamr network with Python.
- Represented the platform at technology events, mentoring students hands-on through the hardware modules.
- Automated repository and release workflows with Git.

Python · Raspberry Pi · ESP32 · Blockchain / IoT · Git

Education

UNAM — Center for Applied Physics and Advanced Technology (CFATA)

GRADUATED MAY 2016

Bachelor of Technology · Querétaro / Mexico City

- Science and engineering framework with specialisations in physics of energy, electronics, programming and robotics.

MITx — Massachusetts Institute of Technology (edX)

2024

6.00.1x — Introduction to Computer Science and Programming Using Python

- Verified certificate, issued by edX.

Codecademy

2021 — 2022

Career paths · Online

- Python 3, Django, JavaScript, React and UI/UX design paths.

Teaching & Workshops

Ironhack México

FEB 2023

Instructor — UX/UI Bootcamp, Web Programming

- Taught a UX/UI bootcamp cohort the fundamentals of web programming, responsive Figma prototyping and a VS Code + GitHub workflow.

Tecnolochicas · Epic Queen · Fundación Televisa · Microsoft

OCT — NOV 2022

Mentor — Web Programming

- Mentored a women-in-tech cohort through HTML, CSS and JavaScript, working in VS Code and GitHub.

EUM Tech · Walmart account

SEP 2022 — JAN 2023

Internal Coding Crash Course — Python & JavaScript

- Trained the engineering team in Python object-oriented programming, nested class models and Django database automation.
- Covered client/server WebSockets across Node.js, Python and ESP32 (Espressif ARM) targets.

Talent Land / Talent Republic · with Streamr

NOV 2020

Speaker — Online IoT Workshop

- Taught an end-to-end IoT pipeline: RUUVI open-source BLE sensors captured with Python on Raspberry Pi, streamed to the Streamr blockchain marketplace and visualised in a Plotly React app.

Microsoft OpenDevCamp · with AMESOL

MAR 2020

Speaker — Azure IoT Workshop · Hotel Galería Plaza Reforma, Mexico City

- Ran an open-source IoT workshop connecting hardware to Azure Cloud, using a Raspberry Pi gateway programmed in Python.

Talent Land

AUG 2019

Speaker — Internet of Things · Guadalajara Convention Center

- Delivered "Percibiendo la realidad con Raspberry Pi", a two-part session on sensing the physical world with Raspberry Pi.

Nehenemi Labs

MAR 2019

Instructor — Raspberry Pi Day · Centro Cultural Gómez Morín

- Taught open-source hardware fundamentals at the community Raspberry Pi Day celebration.

Encuentro Educación Maker México · with Edukreativos

AUG 2018

Instructor — Wearables Workshop · Colegio Hebreo Maguen David

- Taught wearable electronics using ATTINY Atmel microcontrollers to drive a programmable light bracelet.

Creative Engineering

A1 Laboratorio FX

JUL — SEP 2022

Product Design Engineer · Mexico City

- Designed a wearable lighting product fitted inside a character prop skull, built around an ESP32 core with a rechargeable power regulator and selectable light patterns so the art director could change the look on set.
- Programmed an interactive JavaScript invitation for the lab's anniversary.

ESP32 · JavaScript · Power Electronics

Bioscénica · Hierophania Lab · Hemispheric Institute

JUN 2019

Creative Light & Sound Coding · Mexico City

- Coded the light behaviour of the "Suavicrema" façade at Centro de Cultura Digital for *Empatía 5.1*, a site-specific technoshamanic ritual: a wearable ESP32 heartbeat sensor streamed over WebSockets, UDP and OSC into a VVVV light server that turned the audience's pulse into the building's lighting in real time.
- Built a digital sculpture driven by cosmic-ray data, converting the readings into light rendered in a VR headset.

ESP32 · WebSockets · OSC / UDP · VVVV · VR

Technical Skills

LANGUAGES	Python · JavaScript · C++ · MicroPython · SQL · HTML · CSS
FRAMEWORKS & LIBRARIES	Django · Node.js · Express · React · Three.js · jQuery · Flexbox
HARDWARE & EMBEDDED	Raspberry Pi · ESP32 (Espressif) · Atmel / ATTINY · Ultra-Wideband RF · BLE sensors · Serial & WebSocket protocols · Arduino IDE
TOOLS & PLATFORMS	Git & GitHub · VS Code · Jupyter · Conda · npm · Webpack · Jira · Microsoft Power Apps · SharePoint · Azure · Apache · SSH · CLI / Shell
ELECTRONICS & DESIGN	KiCad · Eagle · EasyEDA · Figma · Adobe Illustrator · Adobe Photoshop · Spline · Blender · Unity
PRACTICES	Agile · Full SDLC · Clean Code & SOLID · PEP-8 · Code review · Technical documentation · UI/UX prototyping

Languages

Spanish Native · **English** Professional working proficiency

Awards & Recognition

Speaker — IoT with Python & Azure

OpenDevCamp / Microsoft · Hotel Galería Plaza Reforma, 2020

Community Engineer Hero

Streamr · Talent Land Jalisco, April 2019

Mentor — MXHACKS V

Major League Hacking Hackathon, 2018

Bots Crash Course

Wizeline Academy · Centro, 2017

Facebook Hackathon

Aldea Digital 4G · Mexico City, 2017

Eric Margay · Hardware & Software Engineer · Mexico City